

SMART GRADE

Features Sheet

SIN GBHS FOUMBAN — Form 5B Science
Academic Year 2026–2027

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1. Overview

SMART GRADE is a complete Progressive Web Application (PWA) for grade management, working 100% offline after first installation, with all data stored locally on the student's device.

Subjects	14
Sequences	6 (2 per term)
Terms	3
Badges	23
Themes	20
Fonts	12
Search Sources	15
AI Models	3

2. Academic Features

Feature	Description
14 Subjects	Customizable coefficients (1-10)
6 Sequences	2 per term, 3 terms per year
Smart Rounding	Automatic compensation system
Advanced Stats	Predictions and performance analysis
Yearly Report	Letter grades (A+, A, B+, B, C, D, F)

Analytics and Visualization

Feature	Description
Bar Charts	Interactive Chart.js graphs
Radar Charts	Subject comparison
Doughnut Charts	Grade distribution
Zoomable	Progress chart with zoom
Trend Analysis	Performance over time

Study Tools

Feature	Description
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Flashcards	Default + custom cards per subject
Goals System	Progress bars for targets
Timetable	Class schedule viewer
23 Badges	Achievement and streak badges
Streak Tracking	Daily usage tracking

3. Security

Feature	Description
Fingerprint	WebAuthn API biometric login
4-digit PIN	Secure authentication
Local Storage	No external servers

4. Export and Transfer

Feature	Description
JSON	Complete backup
CSV	Spreadsheet format
HTML	Viewable report
PDF	Professional report card
Import	JSON backup restore
QR Code	Data transfer
6-digit Code	Local transfer

5. AI Assistant

Feature	Description
Groq	Ultra fast model
Devstral	Medium model
Codestral	Code-focused model
LaTeX	Math support

Prism.js	Code highlighting
Mermaid	Diagram generation
Conversation History	Save and view past conversations

6. Multi-Search Engine

15 sources queried simultaneously for every search:

Feature	Description
DuckDuckGo	General search
Wikipedia	Encyclopedia
Wiktionary	Dictionary
OpenLibrary	Books
WikiMedia	Media
Wikidata	Knowledge base
Google Books	Books
Internet Archive	Archive
Jisho	Japanese dictionary
TheAudioDB	Music database
OMDb	Movies database
NewsAPI	News
YouTube	Videos
Reddit	Discussions
GitHub	Code repositories

7. Notebook

Feature	Description
Personal Notes	Create and manage notes
Pinning	Pin important notes to top
Color Coding	Organize notes by color
Subject Tagging	Tag notes by subject

Full-Text Search	Search through all notes
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8. Homework Manager

Feature	Description
Admin Panel	Teachers can manage assignments
File Attachments	Images, videos, PDFs
Cloud Sync	Via GitHub Gist
Student View	View and track assignments

9. Chat System

Feature	Description
P2P Chat	Via JSONBin.io
Real-Time Sync	Automatic updates
Unique Codes	Each user has a unique code

10. Developer Tools

Feature	Description
Grade Calculator	Advanced calculation tools
Database Manager	View and manage user data
Advanced Statistics	Detailed performance analysis
System Logs	Real activity history
Cloud Backup	Real backup and restore
System Config	View and modify settings

11. Badges System (23 Badges)

Streak Badges (Auto-unlock)

Feature	Description
Beginner Streak	3 consecutive days

Regular Streak	7 consecutive days
Dedicated Streak	15 consecutive days
Legendary Streak	30 consecutive days

Achievement Badges (selection)

Feature	Description
First Grade	Add your very first grade
Perfect Score	Get 20/20 in any subject
High Average	Overall average 12/20 and above
Bookworm	Record 10+ grades
Dedication	Record 30+ grades
Scholar	Unlock 8 achievements
Photo Uploader	Add a profile photo
Theme Collector	Try 10 different themes
Font Collector	Try 6 different fonts

12. Themes and Fonts

20 color themes are available, along with 12 font families (Inter, Roboto, Cinzel, Quicksand, Courier Prime, Fredoka, Pacifico, Bangers, Lobster, Permanent Marker, Comfortaa, Righteous).

Night mode activates automatically between 8 PM and 6 AM, and can also be toggled manually at any time.

13. Technology Stack

Feature	Description
HTML5	Semantic markup, LocalStorage, Canvas, MediaDevices, Notifications
CSS3	Flexbox, Grid, CSS variables, glassmorphism, media queries
JavaScript ES6+	Promises, Async/Await, LocalStorage database management
Chart.js	Interactive charts
html2pdf.js	PDF generation
QRCode.js / html5-qrcode	QR code generation and scanning
Service Worker	Offline caching

Web App Manifest	Installation support
WebAuthn API	Biometric authentication

Contact

Feature	Description
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